

# Final report of photonic integrated circuit design

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## 1. Introduction

Photonic integrated circuits have a wide range of applications in biosensing [1], optical communication [2], and in quantum technologies including creating entangled photons, quantum state manipulation [3,4], and beam delivery for trapped atom experiments [5–7]. Hence, it is currently an extremely active area of research and has been for the last 20 years.

The Mach-Zehnder interferometer is a circuit that has been employed in many of these applications, as a sensor [8], a key component in quantum state preparation [9], and for tuning the properties of the light in the photonics integrated circuit. Through the design and fabrication of a range of Mach-Zehnder interferometers, greater understanding of the process and the experimental limitations and variations will be achieved. An application that is of particular interest in this study is the creation of switches with Mach-Zehnder interferometers and Bragg gratings for wavelength dependent mirrors in photonic integrated circuits [10, 11].

## 2. Theoretical background

In this section, theory and expressions required for analysing the simulated and experimental results of Mach-Zehnder interferometers and Bragg gratings are introduced.

### 2.1. Strip waveguide

The strip waveguide is used to create the circuits in this study, which has a rectangular cross section with variable parameters width,  $w$ , and thickness,  $h$ , that determine the modes of propagation in the waveguide. An effective waveguide should contain all of the extent of the electric and magnetic field, guiding the light in the desired direction [12]. A silicon waveguide is depicted in Figure 1.

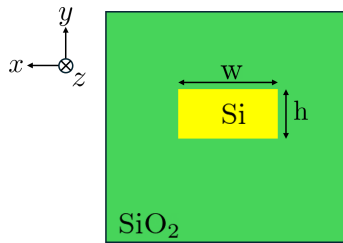


Fig. 1. Cross section of a silicon waveguide, where the waveguide is shown in yellow, made of silicon, and the cladding in green is silicon oxide. The width,  $w$ , and thickness,  $h$ , of the waveguide are labelled.

The propagation of light in the waveguide is in the  $z$ -direction, which is into the page. The width and thickness of the waveguide should be suitable to ensure that propagation through the waveguide is single mode, typically the thickness should be less than 250 nm [2, 13]. Alignment of the electric field in the  $x$ -direction or horizontally is called the transverse electric (TE) mode, and the electric field in the  $y$ -direction or vertical is the transverse magnetic (TM) mode. In reality, waveguides do not only support the fundamental TE<sub>0</sub> and TM<sub>0</sub> modes, therefore a more accurate description is quasi-TE and quasi-TM.

The effective index takes into account that part of the electric field propagates is exponentially

decaying in the cladding, and is an important parameter in characterisation of photonic integrated circuits. The compact model for the waveguide is derived from the Taylor expansion around the central wavelength  $\lambda_0$  [13],

$$n_{\text{eff}}(\lambda) = n_0 + n_1 (\lambda - \lambda_0) + n_2 (\lambda - \lambda_0)^2, \quad (1)$$

where  $n_0, n_1, n_2$  are the amplitudes of the respective polynomial orders, which will be modelled in Section 3.1 for the specific waveguide materials and dimensions considered.

## 2.2. The Mach-Zehnder interferometer

The Mach-Zehnder interferometer (MZI) is a simple design in integrated photonics, comprising two Y-branches with a length of waveguide in between [14], as depicted in Figure 2. The figure shows the imbalanced interferometer case where there is an extra length of waveguide in arm 2 creating a path length difference.

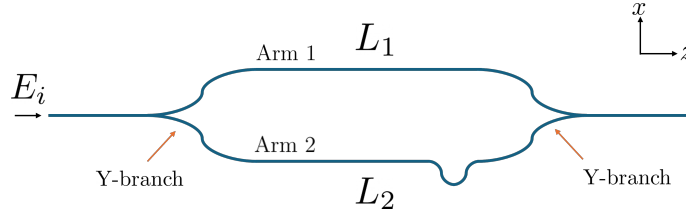


Fig. 2. Diagram of a Mach-Zehnder interferometer with input plane wave,  $E_i$ . The path lengths  $L_1$  and  $L_2$  are not equal, shown with the loop in arm 2 of the interferometer.

The ratio of the input intensity to output intensity of the interferometer is an important characteristic called the transfer function, which is derived as follows. Consider the lossless case of propagation through the interferometer. The Y-branch splits the input plane wave in two,

$$E_i = E_0 e^{i(\omega t - \beta z)} \quad (2)$$

where  $E_0$  is the electric field amplitude,  $\omega$  is the angular frequency of the light, and  $\beta$  is the propagation constant depending on the effective index,  $n_{\text{eff}}$ , and the wavelength of the light,  $\lambda$ ,

$$\beta = \frac{2\pi n_{\text{eff}}}{\lambda}. \quad (3)$$

Hence the outputs of the Y-branch splitter are

$$E_1 = \frac{E_i}{\sqrt{2}}, \quad E_2 = \frac{E_i}{\sqrt{2}}, \quad (4)$$

where  $E_1$  and  $E_2$  are the electric fields in arm 1 and 2 respectively. After propagation through the waveguide and at the input of the second Y-branch, the electric field in arm 1 is

$$E_{o1} = E_1 e^{-i\beta L_1}, \quad (5)$$

and in arm 2, where  $L_2 = L_1 + \Delta L$ ,

$$E_{o2} = E_2 e^{-i\beta L_2}. \quad (6)$$

The output of the combining Y-branch is described by

$$E_o = \frac{1}{\sqrt{2}} (E_{o1} + E_{o2}) = \frac{E_i}{2} (e^{-i\beta L_1} + e^{-i\beta L_2}) \quad (7)$$

and the associated intensity is

$$I_o = \frac{I_i}{4} |e^{-i\beta L_1} + e^{-i\beta L_2}|^2, \quad (8)$$

where the initial intensity is  $I_i = |E_i|^2$ . This can be simplified further for the imbalanced interferometer used in this study. It is assumed that the effective index and temperature are equal in the two arms, but the path lengths are different. Using trigonometric identities, this can be rewritten as

$$I_o = \frac{I_i}{2} [1 + \cos(\beta\Delta L)] \quad (9)$$

where  $\Delta L = L_1 - L_2$ . Hence, the transfer function of the imbalanced MZI is

$$T_{\text{MZI}} = \frac{I_o}{I_i} = \frac{1}{2} [1 + \cos(\beta\Delta L)]. \quad (10)$$

The sinusoidal nature of the output intensity is free spectral range-like. Using approximations for the dependence of  $\beta$  in wavelength, the free spectral range (FSR) [14], or the separation in adjacent peaks in the transfer function, is

$$\text{FSR} = \Delta\lambda = \frac{\lambda^2}{\Delta L n_g(\lambda)}, \quad (11)$$

where  $n_g$  is the group index [13] that is defined as

$$n_g(\lambda) = n_{\text{eff}}(\lambda) - \lambda \frac{dn_{\text{eff}}}{d\lambda}. \quad (12)$$

In the next section, the FSR will be predicted from the transfer function for a range of path differences, and TM and TE polarisations. A specific setup of a MZI will also be designed with  $2 \times 2$  directional couplers rather than  $1 \times 2$  Y-branches. Directional couplers can be designed to have arbitrary splitting ratios depending on the separation distance and interaction length [15, 16]. When used in the MZI setup, the circuit is opened up to many more applications in quantum technology and quantum computing [9].

### 2.3. Bragg gratings

A Bragg grating is comprised of a waveguide with periodic refractive index variations parallel to the direction of propagation, which creates a one-dimensional photonic bandgap affecting the transmission and reflection spectrums depending on the input wavelength [11]. This behaviour can be applied to create filters, mirrors, and, with multiple Bragg gratings, Fabry-Perot cavities with guided light. Figure 3 shows the ideal spectra transmitted and reflected from a fibre Bragg grating, and although it does not depict a photonic integrated Bragg grating, the ideal response is the same from the two structures. Light is transmitted unaffected through the Bragg grating unless the light is at the resonant wavelength, or close to the resonant wavelength within some bandwidth. When this condition is met, the reflection at each change in refractive index interferes constructively, and a peak in reflection is observed. The bandwidth of the Bragg grating depends on the coupling coefficient,  $\kappa$ , which quantifies this constructive interference.

One method of creating Bragg gratings in photonic integrated circuits is with corrugations in the sidewalls of the waveguide, resulting in changes in the effective index along the waveguide. The main parameter used to control the wavelength of maximum reflection is the modulation period, through the following relation:

$$\lambda_{\text{Bragg}} = \frac{2\Lambda n_{\text{eff}}}{m}, \quad (13)$$

where  $\lambda_{\text{Bragg}}$  is the so-named Bragg wavelength,  $n_{\text{eff}}$  is the effective index of the original unmodulated waveguide,  $\Lambda$  is the refractive index modulation period, and  $m$  is the order of resonance, or grating order. The coupling coefficient and hence bandwidth in a photonic integrated Bragg grating depends on the corrugation width, and the alignment and smoothing of the modulations on the walls of the waveguide [17, 18].

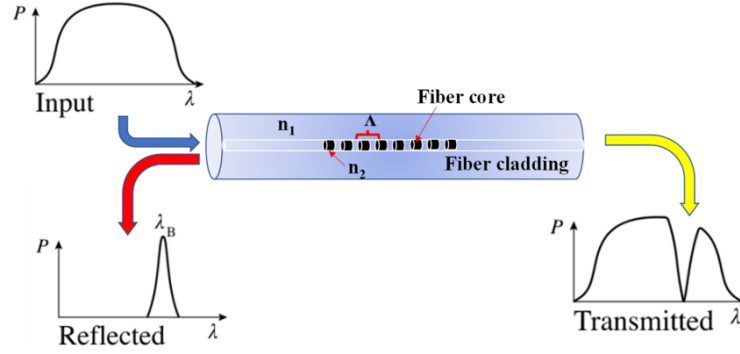


Fig. 3. Depiction of the ideal spectra at input, transmission and reflection through a fibre Bragg grating. The modulation period  $\Lambda$  and refractive index of the cladding  $n_1$  and core  $n_2$ . Figure from [11].

### 3. Modelling and simulation

#### 3.1. Strip waveguide modelling

All of the waveguides used in this study are silicon, 500  $\mu\text{m}$  wide and 220  $\mu\text{m}$  thick, and the cladding is silicon dioxide, as described in Section 2.1. Using Lumerical MODE modelling software, the quasi-TE and quasi-TM fundamental mode profiles the waveguide at 1550 nm were plotted and are shown in Figures 4 and 5 respectively.

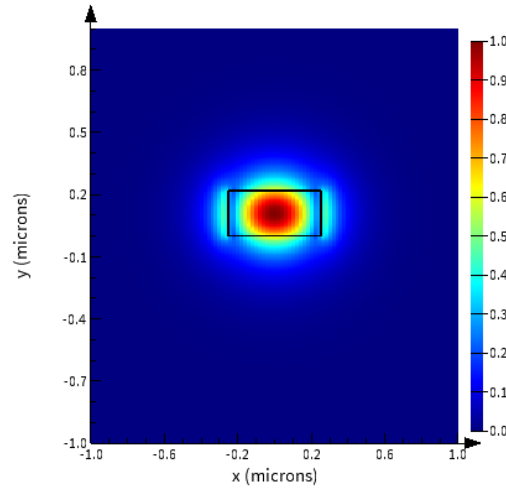


Fig. 4. 2D plot of the  $x$ -component of the electric field, showing this is the fundamental quasi-transverse electric mode in the strip waveguide. The colour bar is the normalised electric field amplitude.

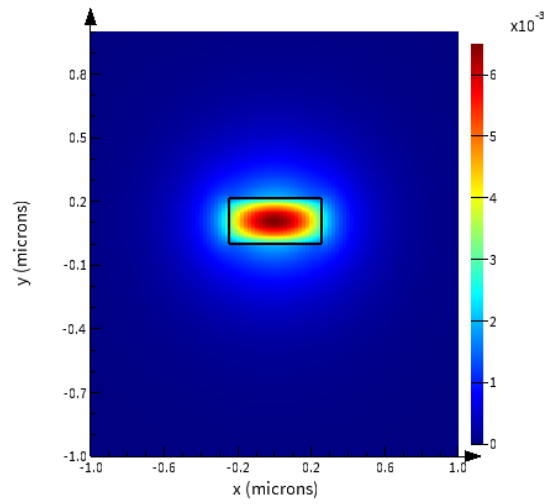


Fig. 5. 2D plot of the  $x$ -component of the magnetic field, showing this is the fundamental quasi- transverse magnetic mode in the strip waveguide. The colour bar is the normalised magnetic field amplitude compared to the electric field amplitude, hence orders of magnitude smaller than in Figure 4.

There is a significant amplitude in cladding for the quasi-TE case in Figure 4, beyond the black box showing the extent of the waveguide. The effective index as a function of wavelength for the quasi-TE mode from Figure 4 is shown in Figure 6.

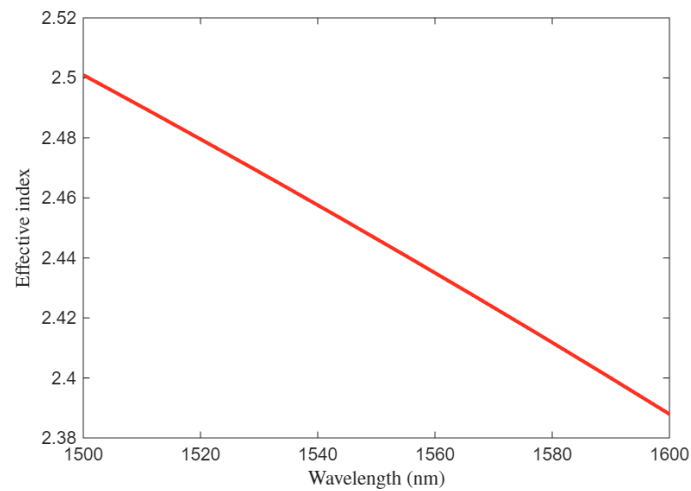


Fig. 6. The simulated effective index for the fundamental quasi-transverse electric mode in a silicon strip waveguide as a function of wavelength.

Fitting to these results with a Lumerical script, the compact model of the waveguide, in the form of Equation 1, is

$$n_{\text{eff}}(\lambda) = 2.44 - 1.13 (\lambda - \lambda_0) - 0.04 (\lambda - \lambda_0)^2, \quad (14)$$

where  $\lambda_0$  is the central wavelength, which is 1550 nm.

For further insight into the quasi-TE mode in the waveguide, the group index as a function of wavelength is also simulated in Lumerical and plotted in Figure 7.

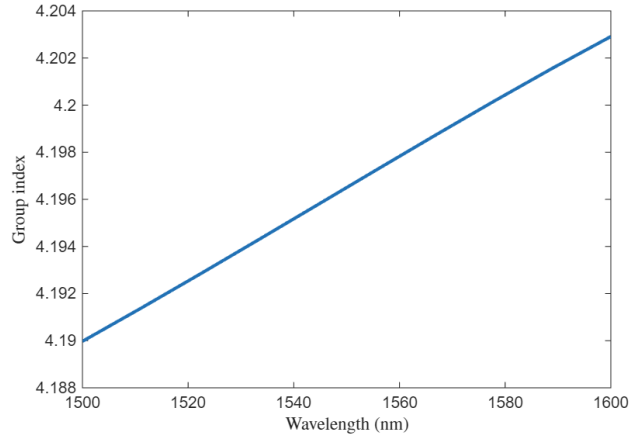


Fig. 7. The simulated group index for the fundamental quasi-transverse electric mode in a silicon strip waveguide as a function of wavelength.

The group index is predicted to vary a small amount over a 100 nm wavelength range, < 3 %. The effective index and group index are both smaller for the quasi-TM mode through the waveguide, where the effective index is simulated to be 1.77 at 1550 nm and the group index is 3.748.

### 3.2. Mach-Zehnder interferometer simulations

Descriptions of the different MZIs in polarisation, path difference  $\Delta L$ , and type of coupler designed and simulated are summarised in Table 1. The MZIs are made with 50:50 Y-couplers except for the bdc MZI which uses 50:50 broadband directional couplers (bdc).

Table 1. The Mach-Zehnder interferometer labels and differences in path length.

| Label    | $L_1$ ( $\mu\text{m}$ ) | $L_2$ ( $\mu\text{m}$ ) | $\Delta L$ ( $\mu\text{m}$ ) |
|----------|-------------------------|-------------------------|------------------------------|
| TE MZI 1 | 100                     | 175                     | 75                           |
| TE MZI 2 | 100                     | 200                     | 100                          |
| TE MZI 3 | 100                     | 300                     | 200                          |
| TE MZI 4 | 100                     | 400                     | 300                          |
| TE MZI 5 | 100                     | 500                     | 400                          |
| TM MZI 1 | 100                     | 300                     | 200                          |
| bdc MZI  | 100                     | 200                     | 100                          |

The TE MZIs form the parameter scan in path difference and one imbalanced interferometer with quasi-TM mode is tested to be compared to the quasi-TE interferometer. It is expected that differences in the bandwidth and loss through the bending waveguides will be observed due to the polarisation and the efficiency of the grating couplers for each polarisation. In addition, the FSR for each polarisation will be similar but not exactly the same due to differences in the group index. TE MZI 3 has an expected FSR of 2.86 nm whereas TM MZI 1 is expected to be 3.16 nm. Finally, an interferometer labelled dbc MZI is an interferometer created with broadband directional couplers rather than Y-branches. Variations can be studied comparing the type of coupler used in the photonics integrated circuit, and suggest any benefits or limitations that arise from the choice.

All of the MZIs in Table 1 were simulated in Lumerical interconnect and the TE MZIs were compared directly. Additional losses are modelled in the simulation to provide a more realistic view of the gain profile as a function of wavelength. The equation for FSR provided in Equation 11 does not account for losses in the waveguide, that arise from roughness in the sidewalls of the waveguide and absorption, so an effective loss is added in the simulations equal to 3 dB/cm. S-parameters for the Y-branches forming the MZI, as shown in Figure 2, are used to model a realistic Y-branch, rather than using the ideal case from the software library. Grating couplers for the corresponding polarisation are simulated as inputs and outputs of the optical network analyser, and hence there is a trend in the gain profile from the insertion loss rather than the gain peaks being a constant power across the wavelength range considered.

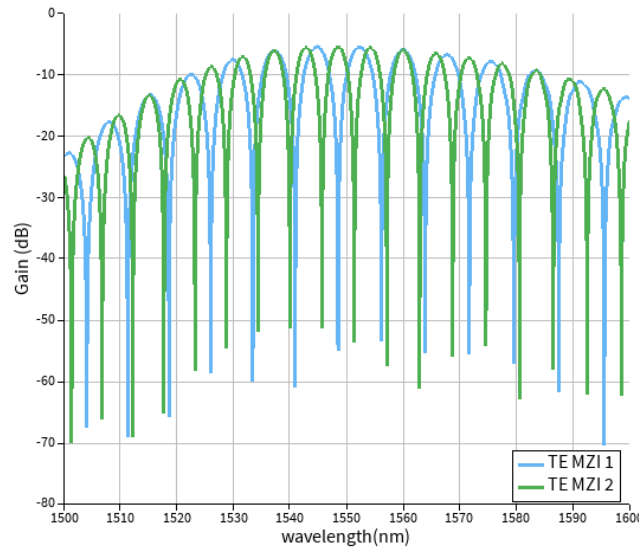


Fig. 8. Simulations of the gain profile as a function of wavelength for two transverse electric (TE) Mach-Zehnder interferometers, as identified in Table 1.

The sinusoidal variation in the gain with varying wavelength is shown in Figure 8 for the two shortest path differences designed. For the longer path difference, TE MZI 2, the FSR, or the separation of adjacent peaks in wavelength, decreases. This inversely proportional trend between FSR and path difference is an important relation and with the parameters chosen the trend is clearly seen in the simulated FSRs, shown in Figure 9.

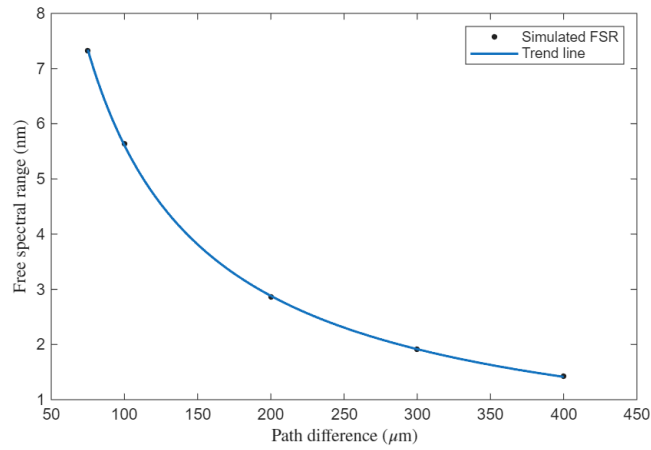


Fig. 9. Free spectral range dependence on the path difference between arms in the Mach-Zehnder interferometer. The data points are for the simulated free spectral range at  $\sim 1550$  nm from Figure 8, and the trend line to guide the eye comes from Equation 11.

Finally, the bdc MZI was designed to have two outputs after the arms recombine in order to study the interference effect in the interferometer. The simulated outputs from the bdc MZI are shown in Figure 10, focusing on a portion of the wavelength scan to see the differences between the outputs clearly.

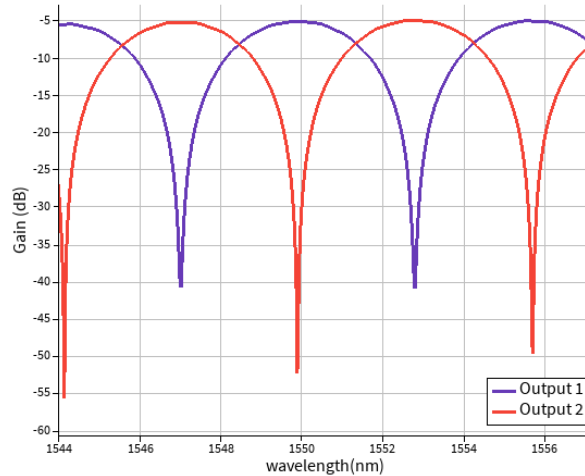


Fig. 10. Simulated gain profile as a function of wavelength for dbc MZI from Table 1; the Mach-Zehnder interferometer formed with directional couplers.

When output 1 in Figure 10 is at its maximum then output 2 is at its minimum, and vice versa, because each output shows constructive and destructive interference of the light in the two arms of the interferometer. This graph shows the motivation for using MZIs to form switches, because for a certain wavelength of input light there is transmission through one of the outputs only. In practise, a balanced MZI with variable phase changers or a heater on one of the arms would be

more suitable for flipping the switch than varying the wavelength for an imbalanced MZI.

### 3.3. Bragg grating simulations

A simple Bragg grating of 300 periods with a modulation period  $\Lambda$  of 317 nm and modulation width of 40 nm on a strip waveguide is also studied. In Lumerical interconnect, the resonant wavelength or Bragg wavelength described in Equation 13 is predicted to be 1548 nm. In addition, the reflection and transmission spectra through the Bragg grating were simulated, and are shown in Figure 11.

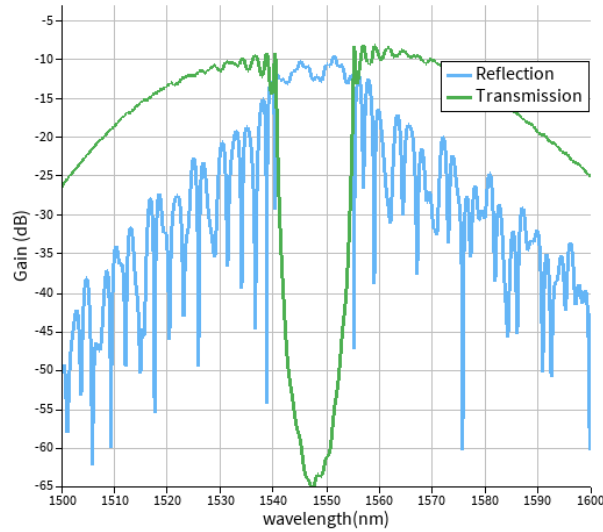


Fig. 11. Simulation of the gain profiles as a function of wavelength for the Bragg grating under study, showing the transmission and reflection components.

The figure shows that the expected bandwidth is 15 nm, and that within this bandwidth, the transmission gain is much less than the reflection gain with an extinction ratio of over 20. Again, the grating couplers creates an envelope for both spectra.

## 4. Fabrication

The structures described and simulated in Section 3, as well as loopback structures of TE and TM modes for removing the background of the experimental data, were created in KLayout in an area of  $605 \mu\text{m} \times 410 \mu\text{m}$ . The layout is shown in the appendix. The SOI (silicon-on-insulator) photonic integrated chip was fabricated by Applied Nanotools Inc. with 100 keV electron-beam lithography [19], which has a minimum feature size of 60 nm. Experimental variation in the fabrication through electron-beam lithography means that a waveguide designed to be 500 nm wide is expected to have a measured size of 470 nm to 510 nm. Likewise, the thickness is expected to be between 215.3 nm and 223.1 nm for a design of 220 nm thick. With a range of dimensions expected, a corner analysis was performed to estimate the variation in the properties of the waveguide and the devices they comprise. Variations in the thickness and width affect the effective index and group index of the TE and TM modes. The differences in effective index and group index were simulated and plotted in Figures 12 and 13.

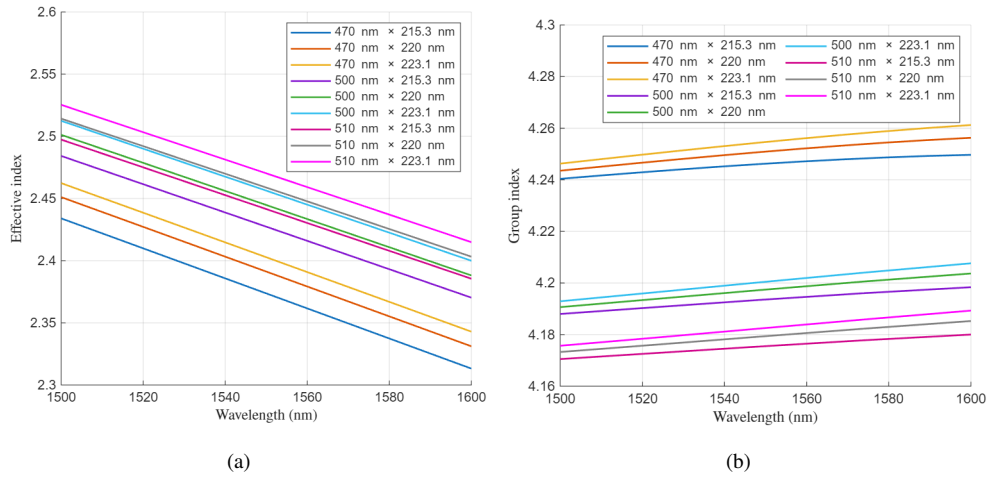


Fig. 12. Corner analysis of a silicon strip waveguide guiding the transverse electric mode. (a) shows the variation in the effective index and (b) the variation in the group index.

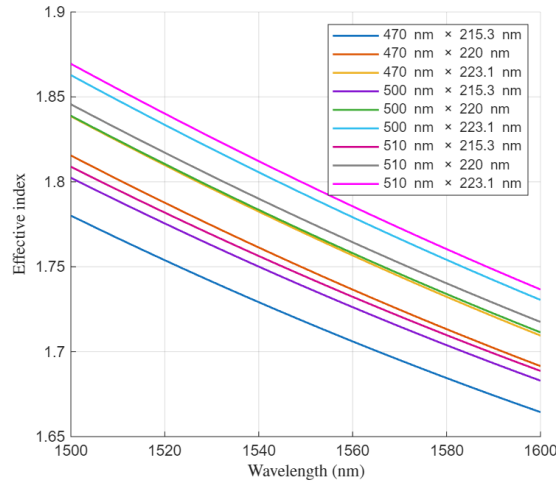


Fig. 13. Corner analysis of the effective index for a silicon strip waveguide guiding the transverse magnetic mode.

The group index for the TM mode varies between 3.51 and 3.86 at 1550 nm. As a result of changes in the effective index, the positions of the peaks in the MZI transfer function shift in wavelength. Figure 14 shows simulated gain profiles for an MZI with path difference 100  $\mu\text{m}$  for various waveguide dimensions in the corner analysis.

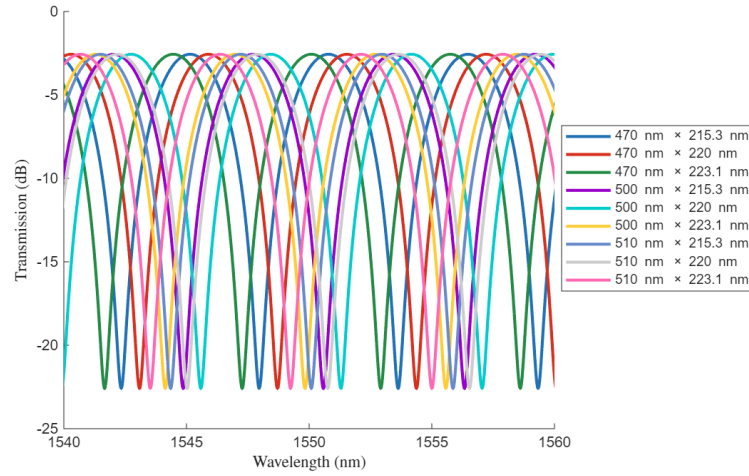


Fig. 14. Variation of the position of the peaks in the transfer function of the Mach-Zehnder interferometer, due to differences in the effective index of the waveguide. Zoomed in on the target wavelength of 1550 nm.

Variations in the group index affect the FSR due to the relation shown in Equation 11. Lower and upper bounds on the FSR at 1550 nm for each MZI fabricated were estimated from the corner analysis group index and these bounds are summarised in Table 2 to be compared to the experimentally extracted FSRs.

Table 2. Upper and lower bound expectations of free spectral range (FSR) from corner analysis for the range of Mach-Zehnder interferometers at 1550 nm.

| Label              | Lower bound FSR (nm) | Upper bound FSR (nm) |
|--------------------|----------------------|----------------------|
| TE MZI 1           | 7.53                 | 7.67                 |
| TE MZI 2 & bdc MZI | 5.65                 | 5.75                 |
| TE MZI 3           | 2.82                 | 2.88                 |
| TE MZI 4           | 1.88                 | 1.92                 |
| TE MZI 5           | 1.41                 | 1.44                 |
| TM MZI 1           | 3.11                 | 3.42                 |

In addition to the change in FSR, variation in the width of the waveguide can affect the splitting ratio of the directional couplers. This effect is not expected to be significant, causing changes in the splitting ratio of a few %, which may be visible as differences in the maximum transmission of each output. Changes in the Bragg grating corrugation, such as becoming sinusoidal instead of rectangular, change the periodic effective index in the grating and the properties of the transmission and reflection. This is equivalent to reducing the width of the corrugation and causes the resonance of the Bragg grating peak to shift to shorter wavelengths [20]. The Bragg grating feature size is smaller than the minimum feature size of electron-beam lithography, so this is expected to have an effect on the reflection and transmission spectra.

## 5. Experimental data

### 5.1. Analysis of Mach-Zehnder interferometers

The photonic chips were probed and measured at the University of British Columbia with an automated four-fibre array setup of polarisation maintaining fibres [13, 21]. A tuneable laser (Agilent 81600B) was used to probe the photonic integrated circuits, with wavelength from 1500 nm to 1600 nm in 10 pm steps. Fits were applied to the TE MZI gain profiles after removing the background with the TE loopback structure, and from this the FSR was extracted. Since the path difference and wavelength is known for each data point, the group index for each MZI was also estimated by rearranging Equation 11. Plots of the raw data and fits for the loopback structure output and TE MZI 1 are shown in the appendix.

The peak wavelength was expected to be 1550 nm due to the envelope of the grating coupler but from experimental data it was  $\sim 1532$  nm. In order to compare to the simulations, the FSR and group index were considered at 1550 nm and not at the experimental central wavelength. The five TE MZIs with varying path differences were analysed, where all of the fits applied had a goodness of fit  $r^2 > 0.989$ , and hence the error from the fit was negligible. The FSRs are quoted to 2 decimal places in Table 3, but without an error estimate since it is limited by the step size of the laser.

Table 3. Experimental results of the free spectral range (FSR) of the transverse electric mode Mach-Zehnder interferometers (MZI) at 1550 nm.

| Label    | FSR (nm) | Group index |
|----------|----------|-------------|
| TE MZI 1 | 7.58     | 4.195       |
| TE MZI 2 | 5.73     | 4.188       |
| TE MZI 3 | 2.86     | 4.193       |
| TE MZI 4 | 1.91     | 4.190       |
| TE MZI 5 | 1.43     | 4.192       |

Compared to the bounds set by corner analysis in Table 2, all of the measured FSRs are within the expected range, and therefore the MZIs have been well modelled within the known fabrication error. Furthermore, comparing the group index values extracted from the fits to Figure 12b, the spread of group index values is within the corner analysis variation and tightly distributed around the mean of 4.192, with a standard deviation of 0.003.

Initial observations of differences between the TM and TE MZIs are that the bandwidth of the TM MZI is much larger than the TE MZIs, and that there is no noticeable difference in insertion loss, all with a maximum at around -22 dB. The raw data for the TM MZI is plotted in the appendix. Defining the bandwidth as a 10 dB below the maximum insertion loss, the bandwidth of the TM MZI is 80 nm compared to 50 nm for the TE MZIs. This is likely due to the bandwidth of the grating couplers, and the TM grating coupler has a more broadband response. For the TM MZI, the TM loopback structure was not measured, which could have been due to high insertion loss or an error in the fabrication or design, and so the data was flattened by fitting to the envelope of the raw data of the MZI. This worked well and the goodness of fit  $r^2$  was still greater than 0.99. The group index extracted for the TM mode was 3.761, which is also within the simulated group index values from corner analysis. The FSR was found to be  $(3.18 \pm 0.04)$  nm, which also agrees with the bounds on the FSR from the corner analysis. The error in the FSR is estimated from the variation in measured FSR results from the transmission profile around 1550 nm. The bdc MZI has two outputs after interference, as discussed in Section 2.2. The raw data of the two outputs

is plotted in Figure 15 and the data after the background was removed using the TE loopback structure is plotted in Figure 16.

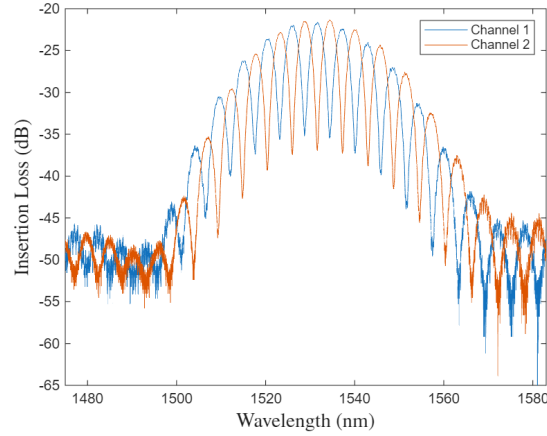


Fig. 15. Raw data from the directional coupler Mach-Zehnder interferometer outputs.

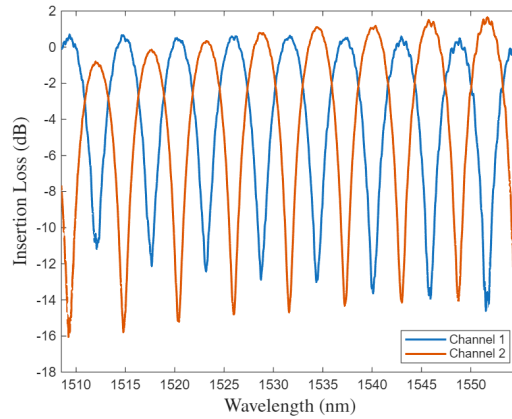


Fig. 16. Directional coupler Mach-Zehnder interferometer outputs. When one output is at the maximum, the other output is a minimum due to the behaviour of the directional coupler.

The extinction ratio between the maximum and minimum of the transfer function of one channel is smaller than the Y-branch MZIs, and the FSR-like sinusoidal oscillations have an envelope structure, which was not observed in the other MZIs. Hence, after the removal of the background with the TE loopback structure, a linear trend is observed in channel 2 in Figure 16 and more noticeably a trend in the peaks in channel 1. This has caused the fit of channel 1 output data at the peaks to not agree near 1550 nm, as seen in Figure 17, but overall the fit is still very good.

The FSR extracted from the fit was found to be  $(5.78 \pm 0.02)$  nm, where the error comes from the differences in the FSR measured for the two channels and different peaks near 1550 nm. This is larger than the predicted FSR from corner analysis, but it is not a significant difference, less than 1 %.

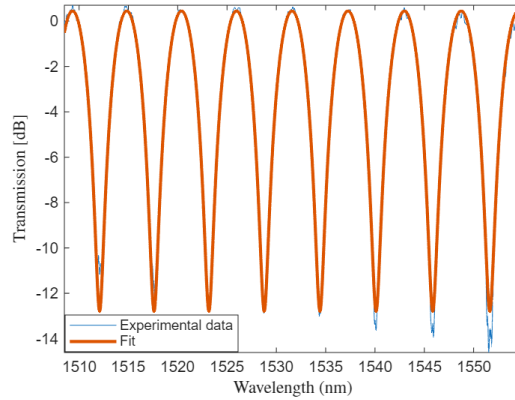


Fig. 17. Fit to experimental data of channel 1 of the directional coupler Mach-Zehnder interferometer, after the background has been removed.

### 5.2. Analysis of the Bragg grating

The raw data of transmission and reflection from the Bragg grating is plotted in Figure 18, where there are a couple of differences observed compared to the simulated spectra in Figure 11.

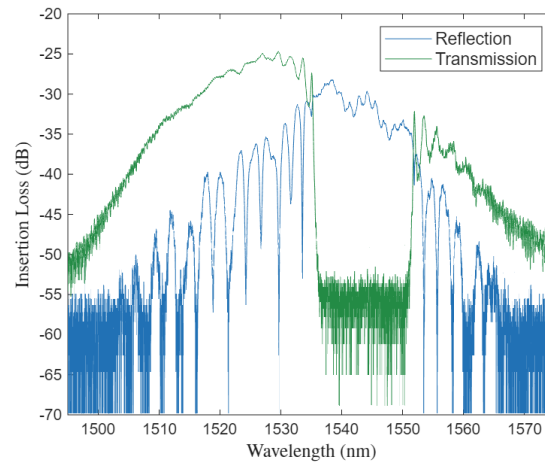


Fig. 18. Measured spectra of the transmission and reflection of the Bragg grating with varying wavelength.

The simulated minimum in transmission at  $-65$  dB is not observed because the transmission goes into the noise floor of the detector at  $-55$  dB. It is also clear from the figure that the grating couplers and the Bragg grating do not have the same resonant wavelength and therefore the envelope of the grating coupler makes the spectra look asymmetric. This did have a detrimental effect on the analysis of the data.

First, the reflection spectrum was analysed to find the Bragg wavelength and the bandwidth. The bandwidth is defined as the range that is less than 3 dB below the maximum power in reflection, and due to the grating coupler creating a negative slope in the reflection peak, this gives an underestimate of 10 nm. The Bragg wavelength is also skewed to a lower wavelength; the peak

in reflection is at 1539 nm compared to the expected 1548 nm, although the difference in these wavelengths is small, less than 1 %. Therefore, the transmission spectrum was also analysed to give a result for the bandwidth that is expected to be more accurate. The bandwidth was measured to be 16 nm, defined at the wavelengths where there is a steep drop off in gain due to enhanced reflection, which agrees closely with the expected bandwidth.

## 6. Conclusion

A range of Mach-Zehnder interferometers with TE and TM polarisation and different coupler components were successfully fabricated and experimentally studied. The free spectral ranges of each MZI were extracted from the experimental data and all of the experimental FSRs agreed with the simulated result within the corner analysis at 1550 nm, except for the MZI made with broadband directional couplers. However, the measured result of  $(5.78 \pm 0.02)$  nm had a relative error of less than 1 % to the upper bound expected result of 5.75 nm. The reason for the difference in FSR compared to the simulation was inappropriate removal of the background of the experimental data. A loopback structure with two outputs and including directional couplers should be considered for future runs in order to properly remove the background and reduce the error. From the FSR, the group index for the TE and TM mode in the strip waveguide were experimentally measured to be  $4.192 \pm 0.003$  and 3.761 respectively, which both agreed with simulations from corner analysis.

In addition to Mach-Zehnder interferometers, a Bragg grating was also studied. The bandwidth of the Bragg grating was measured to be 16 nm, which agrees with the expected bandwidth of 15 nm. The Bragg wavelength was measured to be 1539 nm, which differed slightly compared to the expected result of 1548 nm, but again within less than 1 %. Using a loopback structure with two outputs could also benefit the Bragg grating experimental data and reduce the effect of trends in the baseline. The bandwidth of the transmission spectrum was taken due to inaccuracies in the reflection spectrum data underestimating the bandwidth to be 10 nm.

## Acknowledgements

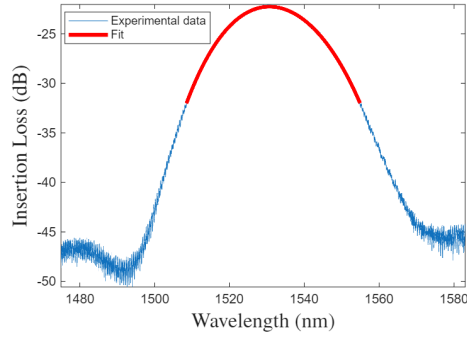
I would like to thank Prof. Lukas Chrostowski and the team at the University of British Columbia for their time spent on the course content, measurements, and leading discussions on the photonics integrated chip topics. I acknowledge the edX UBCx Phot1x Silicon Photonics Design, Fabrication and Data Analysis course, which is supported by the Natural Sciences and Engineering Research Council of Canada (NSERC) Silicon Electronic-Photonic Integrated Circuits (SiEPIC) Program. The devices were fabricated by Richard Bojko at the University of Washington Washington Nanofabrication Facility, part of the National Science Foundation's National Nanotechnology Infrastructure Network (NNIN), and Cameron Horvath at Applied Nanotools, Inc. Omid Esmaeeli performed the measurements at The University of British Columbia. I acknowledge Lumerical Solutions, Inc., Mathworks, and KLayout for the design software.

## References

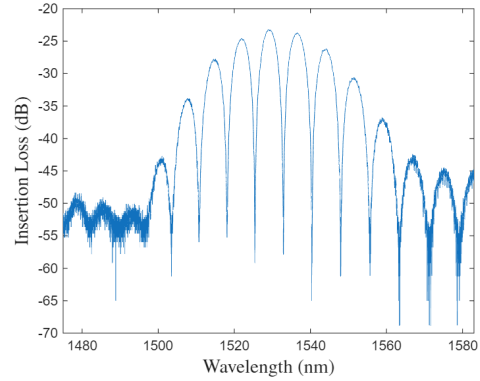
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## Appendices

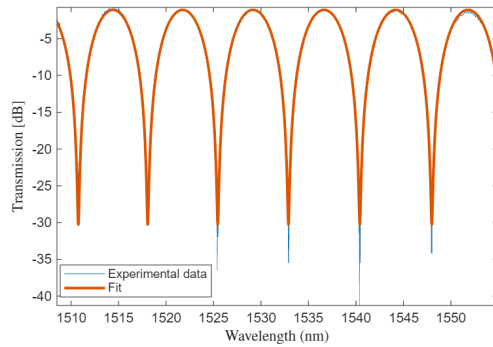
### Data analysis of MZI gain profiles and raw data



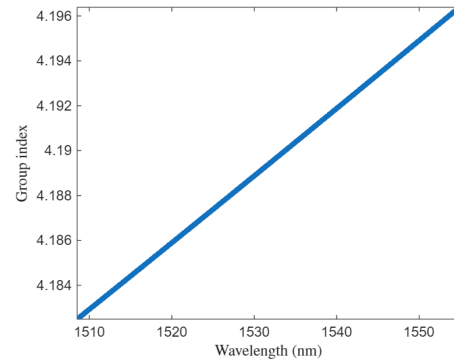
(a) Loopback structure for TE mode, with fit to find the region of interest for the FSR fit.



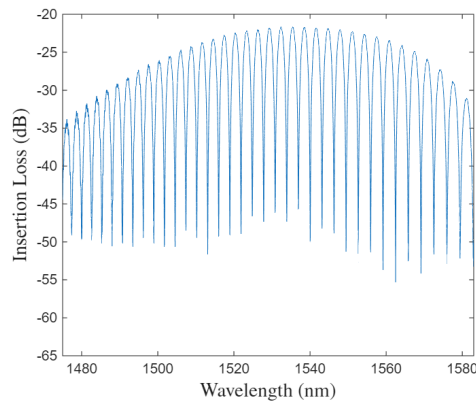
(b) Raw data of TE MZI 1.



(c) Fitted experimental data after the background, Figure a, is removed for TE MZI 1.



(d) Variation of the group index with wavelength extracted from TE MZI 1.



(e) Raw data of TM MZI.

### Photonic chip design

